

Half-Period Involutions and Exact Cancellation in Inert Lucas Sequences

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Abstract

We organize two elementary and exact cancellation mechanisms for quadratic-character sums attached to Lucas sequences modulo an inert odd prime. The first is translation by a matrix half-period, yielding the anti-periodicity relation $U_{n+T/2} = -U_n$. The second is rank-block scalar propagation, equivalently a geometric character factorization, in which the multiplier Λ^k filters complete character sums. These mechanisms arise from different structural viewpoints: the former is an odd-function pairing under index translation, whereas the latter is multiplicative filtering by the rank scalar. The determinant/norm relation links the companion-matrix determinant to the field norm of a characteristic root and forces $\text{ord}(Q) \mid T$; it is kept as an independent observation and is not itself a cancellation criterion. In the quadratic-character full-period setting, the two criteria are nevertheless related: the half-period criterion implies the rank-block scalar criterion, while the converse need not hold. No new rank-period theory is claimed.

Keywords: Lucas sequences; rank of apparition; matrix period; finite-field characters; quadratic characters; exact cancellation.

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1 Introduction

Let $(U_n)_{n \geq 0}$ be the Lucas sequence

$$U_0 = 0, \quad U_1 = 1, \quad U_{n+2} = PU_{n+1} - QU_n, \quad P, Q \in \mathbb{Z}.$$

For an odd prime p , we study complete-period sums of the form

$$\sum_n \chi_p(U_n),$$

where χ_p is the quadratic character of $\mathbb{F}_p$, extended by $\chi_p(0) = 0$.

Periodicity, ranks of apparition, and character sums for Lucas sequences have a substantial classical and modern literature; see, among others, [9, 7, 19, 4, 1, 16, 18, 12, 17, 10, 11, 15, 6, 5, 2, 13, 14, 8]. The purpose of this concise note is structural. We separate two cancellation arguments that are often used in different forms. One is a half-period involution of the companion matrix. The other is rank-block scalar propagation, or equivalently geometric character factorization, arising from the scalar multiplier of each rank block. The determinant/norm relation is kept separate as an independent observation about the matrix period. The distinction matters, especially when $\chi_p(-1) = 1$ but $\chi_p(\Lambda) = -1$. Among recent contributions, Fiebig, Mbirika, and Spilker [5] develop a period-entry-point framework for Lucas sequences modulo m , including structural relations among periods, entry points, and associated orders. The present work

considers a more specialized inert-prime setting, where the companion matrix admits a quadratic-extension interpretation. In this setting, the multiplier associated with the rank of apparition provides an additional scalar structure that leads to exact character-sum factorization. Thus, the present approach complements the period–entry-point viewpoint by exploiting multiplicative-order information available in the inert quadratic-extension framework.

The contribution of the note can be summarized as follows. We formulate two exact cancellation criteria for complete-period character sums in the inert Lucas setting — an index-translation criterion induced by the half-period involution $A^{T/2} = -I$, and a rank-block scalar criterion induced by $A^\alpha = \Lambda I$ — and we prove that, for the quadratic character over a full matrix period, the former implies the latter, while the converse fails as a cancellation argument when $\chi_p(-1) = 1$ ([proposition 5.7](#), [example 6.2](#)). The mechanisms themselves are elementary; the point is their precise separation and the resulting hierarchy.

2 Setup

Throughout, let $P, Q \in \mathbb{Z}$, let p be an odd prime, and put

$$D = P^2 - 4Q.$$

We place the discussion in the standard Lucas-sequence framework by assuming non-degeneracy over characteristic zero. The finite-field arguments of this note rely only on the local conditions

$$p \nmid QD$$

and inertness, and all congruences in what follows are taken modulo p .

We use the following standard terminology.

Definition 2.1. The Lucas sequence is *non-degenerate* if, for its characteristic roots λ and μ in characteristic zero, the quotient λ/μ is not a root of unity. The prime p is called *inert* for the sequence if

$$\chi_p(D) = -1;$$

equivalently, the polynomial

$$f(X) = X^2 - PX + Q \in \mathbb{F}_p[X]$$

is irreducible.

In the inert case,

$$\mathbb{F}_{p^2} = \mathbb{F}_p(\sqrt{D}),$$

and the two roots of f are

$$\lambda = \frac{P + \sqrt{D}}{2}, \quad \mu = \frac{P - \sqrt{D}}{2}.$$

They satisfy

$$\lambda + \mu = P, \quad \lambda\mu = Q, \quad \mu = \lambda^p, \quad \lambda = \mu^p.$$

The Binet formulas are

$$U_n = \frac{\lambda^n - \mu^n}{\lambda - \mu}, \quad V_n = \lambda^n + \mu^n, \tag{1}$$

where

$$V_0 = 2, \quad V_1 = P, \quad V_{n+2} = PV_{n+1} - QV_n.$$

Both sequences are regarded modulo p .

Let

$$A = \begin{pmatrix} P & -Q \\ 1 & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_p).$$

For $n \geq 1$,

$$A^n = \begin{pmatrix} U_{n+1} & -QU_n \\ U_n & -QU_{n-1} \end{pmatrix}. \quad (2)$$

In particular, the lower-left entry of A^n is U_n . This companion-matrix viewpoint is classical; for the Fibonacci matrix modulo m , see [12].

3 Matrix Period

Define

$$T = \mathrm{ord}_{\mathrm{GL}_2(\mathbb{F}_p)}(A).$$

Since $A^T = I$, the sequences $(U_n)_{n \geq 0}$ and $(V_n)_{n \geq 0}$ are periodic modulo p , and T is a common period for both sequences.

Proposition 3.1 (Order of the eigenvalues). *In the inert case,*

$$T = \mathrm{ord}_{\mathbb{F}_{p^2}^\times}(\lambda) = \mathrm{ord}_{\mathbb{F}_{p^2}^\times}(\mu).$$

Consequently,

$$T \mid p^2 - 1.$$

Proof. The matrix A is diagonalizable over $\mathbb{F}_{p^2}$, with eigenvalues λ and μ . Since $\mu = \lambda^p$ and the Frobenius map $x \mapsto x^p$ is an automorphism of $\mathbb{F}_{p^2}^\times$, the two eigenvalues have the same multiplicative order. Hence the order of A is their common order. This order divides

$$|\mathbb{F}_{p^2}^\times| = p^2 - 1.$$

□

Proposition 3.2 (Half-period criterion). *For an integer m , one has*

$$A^m = -I$$

if and only if T is even and

$$m \equiv \frac{T}{2} \pmod{T}.$$

In particular, if T is even, then

$$A^{T/2} = -I \quad \text{and} \quad U_{n+T/2} = -U_n, \quad V_{n+T/2} = -V_n$$

for every nonnegative integer n .

Proof. By [proposition 3.1](#), $A^m = -I$ is equivalent to

$$\lambda^m = \mu^m = -1.$$

Since λ and μ both have order T , this occurs precisely when T is even and

$$m \equiv T/2 \pmod{T}.$$

For $m = T/2$, the identity $A^{T/2} = -I$ and the matrix formula (2) give the assertion for U_n . Alternatively, the Binet formulas (1) give both anti-periodicity relations directly. □

Remark 3.3 (Determinant and norm). Since

$$\det(A) = Q,$$

the relation $A^T = I$ implies

$$Q^T = 1 \pmod{p}, \quad \text{ord}_{\mathbb{F}_p^\times}(Q) \mid T.$$

The same relation has the field-theoretic interpretation

$$\text{Norm}_{\mathbb{F}_{p^2}/\mathbb{F}_p}(\lambda) = \lambda\lambda^p = \lambda\mu = Q.$$

Thus the determinant of the companion matrix is the norm of either characteristic root. This is an independent observation and does not itself imply a half-period involution.

4 Half-Period Translation and Determinant/Norm Observation

The half-period relation in [proposition 3.2](#) is a translation statement on indices. It should not be confused with the determinant or norm constraint in [remark 3.3](#). The latter forces the order of Q to divide the matrix period, but does not by itself imply that the period is even or that its midpoint acts as $-I$.

For a function $g : \mathbb{F}_p \rightarrow \mathbb{C}$ satisfying

$$g(-x) = -g(x),$$

the half-period relation immediately yields exact cancellation on any index set invariant under the translation $n \mapsto n + T/2$.

Proposition 4.1 (Half-period cancellation). *Assume that T is even. Let $I \subseteq \mathbb{Z}/T\mathbb{Z}$ be stable under*

$$n \mapsto n + \frac{T}{2}.$$

If $g : \mathbb{F}_p \rightarrow \mathbb{C}$ is odd, then

$$\sum_{n \in I} g(U_n) = 0 \quad \text{and} \quad \sum_{n \in I} g(V_n) = 0.$$

Proof. The translation $n \mapsto n + T/2$ partitions I into pairs. By [proposition 3.2](#),

$$g(U_{n+T/2}) = g(-U_n) = -g(U_n),$$

and the same argument applies to V_n . Summing over the pairs proves both identities. □

Corollary 4.2 (Quadratic half-period cancellation). *If T is even, I is stable under $n \mapsto n + T/2$, and*

$$\chi_p(-1) = -1,$$

then

$$\sum_{n \in I} \chi_p(U_n) = 0 \quad \text{and} \quad \sum_{n \in I} \chi_p(V_n) = 0.$$

Proof. The extension $\chi_p(0) = 0$ preserves the identity

$$\chi_p(-x) = \chi_p(-1)\chi_p(x) = -\chi_p(x)$$

for every $x \in \mathbb{F}_p$. Apply [proposition 4.1](#). □

5 Rank Blocks and Scalar Character Filtering

Let

$$\alpha = \min\{n \geq 1 : U_n = 0 \pmod{p}\}$$

be the rank of apparition of p , and define

$$\Lambda = U_{\alpha+1} \pmod{p}.$$

In the inert setting the set above is nonempty, so α is well defined: [lemma 5.1](#) below identifies α with the order of $\lambda/\mu \in \mathbb{F}_{p^2}^\times$. (For the existence of ranks of apparition in greater generality, see [\[3\]](#).)

Lemma 5.1 (Rank characterization). *In the inert case,*

$$U_n = 0 \pmod{p} \iff \left(\frac{\lambda}{\mu}\right)^n = 1.$$

Consequently,

$$\alpha = \text{ord}_{\mathbb{F}_{p^2}^\times} \left(\frac{\lambda}{\mu}\right).$$

Proof. Since $\lambda - \mu \neq 0$ in $\mathbb{F}_{p^2}$, the Binet formula gives

$$U_n = 0 \iff \lambda^n - \mu^n = 0 \iff \left(\frac{\lambda}{\mu}\right)^n = 1.$$

The assertion follows from the definition of α . □

Corollary 5.2. *In the inert case, $\alpha \mid p+1$.*

Proof. Since $\mu = \lambda^p$ and $\lambda\mu = Q$, one has $\lambda^{p+1} = \mu^{p+1} = Q$, hence

$$\left(\frac{\lambda}{\mu}\right)^{p+1} = 1.$$

Therefore $\text{ord}_{\mathbb{F}_{p^2}^\times}(\lambda/\mu)$ divides $p+1$, and [lemma 5.1](#) gives $\alpha = \text{ord}_{\mathbb{F}_{p^2}^\times}(\lambda/\mu)$. □

Proposition 5.3 (Rank multiplier and period decomposition). *One has*

$$A^\alpha = \Lambda I \quad \text{and} \quad \lambda^\alpha = \mu^\alpha = \Lambda.$$

If

$$\omega = \text{ord}_{\mathbb{F}_p^\times}(\Lambda),$$

then

$$T = \alpha\omega. \tag{3}$$

Moreover, for every $k \geq 0$ and $1 \leq j \leq \alpha$,

$$A^{k\alpha+j} = \Lambda^k A^j, \quad U_{k\alpha+j} = \Lambda^k U_j, \quad V_{k\alpha+j} = \Lambda^k V_j.$$

Proof. Using $U_\alpha = 0$ in (2),

$$A^\alpha = \begin{pmatrix} U_{\alpha+1} & 0 \\ 0 & -QU_{\alpha-1} \end{pmatrix}.$$

The recurrence at index $\alpha-1$ gives

$$U_{\alpha+1} = PU_\alpha - QU_{\alpha-1} = -QU_{\alpha-1},$$

so $A^\alpha = \Lambda I$. Since A^α has eigenvalues λ^α and μ^α , and equals the scalar matrix ΛI , it follows that $\lambda^\alpha = \mu^\alpha = \Lambda$.

By [lemma 5.1](#), $U_n \equiv 0 \pmod{p}$ if and only if $(\lambda/\mu)^n = 1$, and since $\alpha = \text{ord}(\lambda/\mu)$, this holds precisely when $\alpha \mid n$. Now $A^n = I$ implies $U_n = 0$ by the lower-left entry of (2), hence $\alpha \mid n$; writing $n = k\alpha$,

$$A^n = (A^\alpha)^k = \Lambda^k I,$$

which equals I exactly when $\omega \mid k$. Thus $A^n = I$ precisely when $\alpha\omega \mid n$, and (3) follows.

Finally,

$$A^{k\alpha+j} = (A^\alpha)^k A^j = \Lambda^k A^j,$$

which gives the formula for $U_{k\alpha+j}$ by comparison of lower-left entries, and the formula for $V_{k\alpha+j}$ follows from the Binet formula (1):

$$V_{k\alpha+j} = \lambda^{k\alpha+j} + \mu^{k\alpha+j} = \Lambda^k (\lambda^j + \mu^j) = \Lambda^k V_j.$$

□

Remark 5.4 (Classical context). The identity $A^\alpha = \Lambda I$ and the decomposition (3) represent the inert-case realization of classical relationships among periods, ranks of apparition, and multiplier orders for Fibonacci and Lucas sequences [12, 17, 18, 10, 11]. The divisibility $\alpha \mid p+1$ in [corollary 5.2](#) is classical as well [9, 18, 6]. Taking determinants in $A^\alpha = \Lambda I$ further gives

$$\Lambda^2 = Q^\alpha,$$

which connects the rank multiplier with the determinant/norm relation of [remark 3.3](#).

The period–entry–point framework developed by Fiebig, Mbirika, and Spilker [5] provides a complementary description of period, entry–point, and order phenomena for Lucas sequences modulo m . In the present inert prime setting, the multiplier admits an explicit finite-field interpretation through the characteristic root, and its multiplicative order gives the scalar decomposition underlying the rank-block character factorization developed below.

Proposition 5.5 (Full-period character factorization). *Let ψ be any multiplicative character of $\mathbb{F}_p^\times$, extended by $\psi(0) = 0$. For $X = U$ or $X = V$,*

$$\sum_{n=1}^T \psi(X_n) = \left(\sum_{k=0}^{\omega-1} \psi(\Lambda)^k \right) \left(\sum_{j=1}^{\alpha} \psi(X_j) \right). \quad (4)$$

In particular, if

$$\psi(\Lambda) \neq 1,$$

then

$$\sum_{n=1}^T \psi(U_n) = 0 \quad \text{and} \quad \sum_{n=1}^T \psi(V_n) = 0.$$

Proof. Partition the complete period into the blocks

$$n = k\alpha + j, \quad 0 \leq k \leq \omega - 1, \quad 1 \leq j \leq \alpha.$$

By [proposition 5.3](#),

$$X_{k\alpha+j} = \Lambda^k X_j.$$

Therefore,

$$\psi(X_{k\alpha+j}) = \psi(\Lambda)^k \psi(X_j).$$

Summing first over j and then over k proves (4). If $\psi(\Lambda) \neq 1$, the first factor is a geometric sum over a complete cycle of $\psi(\Lambda)$ and is zero. □

Corollary 5.6 (Quadratic scalar cancellation). *If*

$$\chi_p(\Lambda) = -1,$$

then ω is even and

$$\sum_{n=1}^T \chi_p(U_n) = 0 \quad \text{and} \quad \sum_{n=1}^T \chi_p(V_n) = 0.$$

Proof. The condition $\chi_p(\Lambda) = -1$ implies that Λ has even order (if ω were odd, then $\Lambda = (\Lambda^{(\omega+1)/2})^2$ would be a square), so ω and $T = \alpha\omega$ are even. The vanishing follows from [proposition 5.5](#), since $\chi_p(\Lambda) \neq 1$. $\square$

Proposition 5.7 (Quadratic character of the rank multiplier). *In the inert setting:*

(a) *if α is even, then $\chi_p(\Lambda) = -1$;*

(b) *if $p \equiv 3 \pmod{4}$, then $\chi_p(\Lambda) = -1$ if and only if T is even.*

In particular, whenever the half-period criterion of [corollary 4.2](#) applies to a full period, the rank-block scalar criterion of [corollary 5.6](#) applies as well; the converse need not hold ([example 6.2](#)).

Proof. (a) Put $r = \lambda/\mu$, so that $\text{ord}(r) = \alpha$ by [lemma 5.1](#), and recall that $\lambda^\alpha = \Lambda$ by [proposition 5.3](#). By Euler's criterion, since $\Lambda \in \mathbb{F}_p^\times$,

$$\chi_p(\Lambda) = \Lambda^{(p-1)/2} = \lambda^{\alpha(p-1)/2} = (\lambda^{p-1})^{\alpha/2} = (\mu/\lambda)^{\alpha/2} = r^{-\alpha/2}.$$

Since α is even and $\text{ord}(r) = \alpha$, the element $r^{\alpha/2}$ has order 2 in the cyclic group $\mathbb{F}_{p^2}^\times$, hence equals -1 ; therefore $r^{-\alpha/2} = -1$ and $\chi_p(\Lambda) = -1$.

(b) Suppose first that T is even, so that $\alpha\omega$ is even. If α is even, then $\chi_p(\Lambda) = -1$ by (a). If ω is even, then Λ cannot be a square: otherwise $\omega = \text{ord}(\Lambda)$ would divide the order $(p-1)/2$ of the subgroup of squares of $\mathbb{F}_p^\times$, which is odd when $p \equiv 3 \pmod{4}$, contradicting the evenness of ω . Hence $\chi_p(\Lambda) = -1$.

Conversely, suppose $\chi_p(\Lambda) = -1$. If ω were odd, then $\Lambda = (\Lambda^{(\omega+1)/2})^2$ would be a square, a contradiction; hence ω is even, and so $T = \alpha\omega$ is even.

The final assertion follows: on a full period, [corollary 4.2](#) requires T even and $\chi_p(-1) = -1$, that is, $p \equiv 3 \pmod{4}$, and then $\chi_p(\Lambda) = -1$ by (b), so [corollary 5.6](#) applies. $\square$

Remark 5.8 (Two mechanisms and their relation). The cancellation in [corollary 4.2](#) is a translation pairing:

$$X_{n+T/2} = -X_n.$$

It requires an appropriate index-set symmetry and an odd character, or more generally an odd test function. By contrast, the cancellation in [corollary 5.6](#) is rank-block scalar propagation, equivalently a geometric character factorization:

$$X_{k\alpha+j} = \Lambda^k X_j.$$

It requires $\psi(\Lambda) \neq 1$ and does not require $\psi(-1) = -1$. Thus scalar cancellation can occur even when the half-period negation pairing is ineffective, as in [example 6.2](#).

For the quadratic character on a full matrix period, [proposition 5.7](#) nevertheless shows that the two criteria are related: the half-period criterion implies the scalar criterion, while the converse need not hold. Moreover, the scalar criterion also forces the existence of a half-period involution: indeed, $\chi_p(\Lambda) = -1$ implies that ω is even and hence $T = \alpha\omega$ is even, so that $A^{T/2} = -I$ by [proposition 3.2](#). However, the resulting anti-periodicity produces quadratic-character cancellation only when $\chi_p(-1) = -1$.

6 Examples

Example 6.1 (Fibonacci sequence modulo 7). Take

$$(P, Q, p) = (1, -1, 7).$$

Then

$$D = P^2 - 4Q = 5,$$

which is a nonsquare modulo 7. Hence $X^2 - X - 1$ is irreducible over $\mathbb{F}_7$. The companion matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Direct exponentiation gives

$$A^8 = -I, \quad A^{16} = I,$$

so

$$T = 16, \quad U_{n+8} = -U_n \pmod{7}.$$

The rank data provide an independent cross-check:

$$\alpha = 8 = p + 1, \quad \Lambda = U_9 \equiv -1 \pmod{7}, \quad \omega = \text{ord}_{\mathbb{F}_7^\times}(-1) = 2, \quad T = \alpha\omega = 16.$$

Here $\chi_p(-1) = \chi_p(\Lambda) = -1$, so both mechanisms are active: the half-period translation gives

$$\chi_p(U_{n+8}) = -\chi_p(U_n),$$

as in [corollary 4.2](#), and the same vanishing also follows from [corollary 5.6](#).

Example 6.2 (Inert scalar cancellation). Take

$$(P, Q, p) = (1, 2, 5).$$

Here

$$D = 1 - 8 = -7 \equiv 3 \pmod{5},$$

and 3 is a nonsquare modulo 5. Hence $X^2 - X + 2$ is irreducible over $\mathbb{F}_5$. The initial values are

$$U_0, U_1, U_2, U_3, U_4, U_5, U_6, U_7 = 0, 1, 1, 4, 2, 4, 0, 2 \pmod{5}.$$

The rank and scalar multiplier are

$$\alpha = 6, \quad \Lambda = U_{\alpha+1} = U_7 = 2.$$

Moreover,

$$\chi_p(\Lambda) = \chi_5(2) = -1, \quad \omega = \text{ord}_{\mathbb{F}_5^\times}(2) = 4, \quad T = \alpha\omega = 24.$$

Since $\alpha = 6$ is even, the relation $\chi_p(\Lambda) = -1$ is consistent with [proposition 5.7\(a\)](#). The identity $A^\alpha = \Lambda I$ reads $A^6 = 2I$, whence $A^{12} = -I$: anti-periodicity is present. However,

$$\chi_p(-1) = \chi_5(-1) = 1,$$

so the half-period translation does not itself produce quadratic-character cancellation. Instead, [proposition 5.3](#) gives

$$U_{6k+j} = 2^k U_j \pmod{5}, \quad V_{6k+j} = 2^k V_j \pmod{5}.$$

The four rank blocks have quadratic-character sums

$$3, -3, 3, -3,$$

and the alternating geometric factor $\chi_p(\Lambda)^k = (-1)^k$ makes the total vanish. This is scalar character filtering, not a negation pairing.

Remark 6.3 (Even period alone is not enough). The conditions T even and $A^{T/2} = -I$ alone do not force quadratic-character cancellation. For $(P, Q, p) = (1, 1, 5)$ one has $D \equiv 2 \pmod{5}$ (a nonsquare), rank $\alpha = 3$, multiplier $\Lambda \equiv -1 \equiv 4 \pmod{5}$, order $\omega = 2$, and period $T = 6$; then $A^3 = -I$, but $\chi_5(-1) = \chi_5(\Lambda) = 1$ and

$$\sum_{n=1}^6 \chi_5(U_n) = 4 \neq 0.$$

Together, [example 6.1](#), [example 6.2](#), and this remark realize the three configurations of the two quadratic criteria on a full period that can occur: both active, scalar only, and neither. By [proposition 5.7](#), the fourth configuration—half-period cancellation without scalar cancellation—cannot occur.

	p	$\chi_p(-1)$	$\chi_p(\Lambda)$	Half-period	Scalar
example 6.1	7	-1	-1	active	active
example 6.2	5	1	-1	silent	active
remark 6.3	5	1	1	silent	silent

Table 1: The three configurations of the two quadratic criteria realized by the examples. The fourth formal configuration (half-period active, scalar silent) is excluded by [proposition 5.7](#).

7 Scope Remark

This note does not propose a new rank–period theory. The decomposition into rank blocks follows from the companion-matrix identity, the definition of the rank of apparition, and the divisibility $\alpha \mid n$ whenever $U_n = 0 \pmod{p}$, which in the inert setting is an immediate consequence of [lemma 5.1](#). The structural point is the separation of two exact cancellation criteria as arguments: half-period translation and rank-block scalar propagation, equivalently geometric character factorization. For the quadratic character on a full period these criteria are related as in [proposition 5.7](#), but they remain distinct as proofs.

The hypotheses are essential. If $p \mid Q$, the companion matrix need not be invertible. If $p \mid D$, the characteristic polynomial is not separable in the required form. In either situation, the inert quadratic-extension argument used here does not apply without modification. Outside the inert setting considered here, an even matrix period alone does not in general imply that the half-period matrix is $-I$; within the inert setting, T is even if and only if $A^{T/2} = -I$, by [proposition 3.2](#).

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Data Availability

No external datasets were used. The numerical checks and examples are reproducible directly from the displayed recurrences, companion matrices, and finite-field computations; the associated verification code is archived at <https://doi.org/10.5281/zenodo.22118430>. The archived scripts `verify-examples.py` and `verify-theorem-pipeline.py` verify, for the parameter triples displayed in Section 6, the rank of apparition, the multiplier, the matrix period, the half-period identity, and the vanishing of the stated character sums.

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